

Torus and Real-Line Fourier Decomposition Properties Are Equivalent

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Status

This packet records a candidate full solution, likely valid, to Problem 5.2 of Deng, Lorist, and Veraar [1]: whether the Fourier decomposition properties on $\mathbb{T}$ are equivalent to their counterparts on $\mathbb{R}$.

Source Passage

In this section we collect some open problems related to the results of the paper.

The upper and lower decompositions imply (Fourier) type and cotype properties of X as we have seen in Propositions 2.9 and 2.10. It would be interesting to know if a converse result holds.

Problem 5.1. *Let X be a UMD space and $p, q \in (1, \infty)$. Find a sufficient condition for $\ell^q(L^p)$ -upper or lower decompositions in terms of (Fourier) type and cotype of the space X .*

Our decomposition properties are Fourier decomposition properties on $\mathbb{T}$. One may similarly define Fourier decomposition properties on $\mathbb{R}$, in which case it is natural to wonder if these properties would be equivalent. Note that transference methods are not directly applicable.

Problem 5.2. *Are the decomposition properties equivalent to their counterparts on $\mathbb{R}$?*

Transcription of the Open Problem

The source paper writes:

“Our decomposition properties are Fourier decomposition properties on $\mathbb{T}$. One may similarly define Fourier decomposition properties on $\mathbb{R}$, in which case it is natural to wonder if these properties would be equivalent.”

Problem 5.2 asks:

“Are the decomposition properties equivalent to their counterparts on $\mathbb{R}$?”

Definitions Used

Let $G = \mathbb{T}$ or $G = \mathbb{R}$, and let $\widehat{G} = \mathbb{Z}$ or $\widehat{G} = \mathbb{R}$, respectively. For an interval $I \subseteq \widehat{G}$, write

$$D_I^G f = \mathcal{F}_G^{-1}(\mathbf{1}_I \widehat{f}).$$

For a finite interval family $\mathcal{I}$, put

$$A_{\mathcal{I}}^G f = (D_I^G f)_{I \in \mathcal{I}}.$$

The target norm is the mixed norm

$$\|(g_I)_{I \in \mathcal{I}}\|_{\ell^q(L^p(G;X))} = \begin{cases} \left(\sum_{I \in \mathcal{I}} \|g_I\|_{L^p(G;X)}^q \right)^{1/q}, & q < \infty, \\ \sup_{I \in \mathcal{I}} \|g_I\|_{L^p(G;X)}, & q = \infty. \end{cases}$$

Lower $\ell^q(L^p)$ -decompositions mean that

$$\|A_{\mathcal{I}}^G f\|_{\ell^q(L^p(G;X))} \leq L \|f\|_{L^p(G;X)}$$

uniformly over interval partitions and test functions. Upper $\ell^q(L^p)$ -decompositions mean the reverse estimate

$$\|f\|_{L^p(G;X)} \leq U \|A_{\mathcal{I}}^G f\|_{\ell^q(L^p(G;X))}$$

uniformly over interval partitions. As usual, lower estimates may be checked on finite subfamilies. Upper estimates may be checked on finite subfamilies which contain the Fourier support of the test function, since the remaining partition pieces have zero projection. One then passes to general trigonometric polynomials on $\mathbb{T}$, or Schwartz functions with compact Fourier support on $\mathbb{R}$, by density and monotone convergence.

Theorem

Theorem 1. *Let X be a Banach space, let $1 < p < \infty$, and let $1 \leq q \leq \infty$. Then X has lower $\ell^q(L^p)$ -decompositions on $\mathbb{T}$ if and only if it has lower $\ell^q(L^p)$ -decompositions on $\mathbb{R}$. The best constants are equal.*

The same equivalence, again with the same best constants, holds for upper $\ell^q(L^p)$ -decompositions.

Proof Intuition

The decomposition inequalities only involve finitely many interval projections when tested on a polynomial or on a Schwartz function with compact Fourier support. For such a finite family, the map $f \mapsto (D_I f)_I$ is just a finite tuple of Fourier multipliers. De Leeuw’s periodization theorem

compares exactly these multipliers on $\mathbb{R}$ with their sampled versions on $\mathbb{T}$, and de Leeuw's restriction theorem compares a real-line multiplier with its values on integer frequencies.

The apparent obstacle is that the target norm is not a pointwise $L^p(G; \ell^q(X))$ norm, but an ℓ^q -sum of L^p norms. This causes no loss for finite interval families: the standard periodization and restriction proofs give convergence of every component in L^p , and a finite ℓ^q norm commutes with taking the limit. Thus both sides of the decomposition inequalities transfer unchanged.

Lemma 1 (Finite mixed-norm de Leeuw principle). *Let $1 < p < \infty$, $1 \leq q \leq \infty$, and let X be a Banach space.*

1. *Let $\mathcal{J} = \{J_1, \dots, J_N\}$ be a finite family of disjoint intervals in $\mathbb{R}$, and let $f \in \mathcal{S}(\mathbb{R}; X)$ have compact Fourier support whose boundary points do not meet the endpoints of the J_j 's. There are numbers $\varepsilon_n \downarrow 0$, trigonometric polynomials f_n , and finite interval families $\mathcal{J}_n$ in $\mathbb{Z}$ such that*

$$\|f_n\|_{L^p(\mathbb{T}; X)} \rightarrow \|f\|_{L^p(\mathbb{R}; X)}$$

and, for each j ,

$$\left\| D_{J_{j,n}}^{\mathbb{T}} f_n \right\|_{L^p(\mathbb{T}; X)} \rightarrow \left\| D_{J_j}^{\mathbb{R}} f \right\|_{L^p(\mathbb{R}; X)}.$$

2. *Let $\mathcal{I} = \{I_1, \dots, I_N\}$ be a finite family of disjoint intervals in $\mathbb{Z}$, and let $f \in \mathcal{P}(\mathbb{T}; X)$. There are Schwartz functions $F_n \in \mathcal{S}(\mathbb{R}; X)$ and finite interval families $\mathcal{I}_n$ in $\mathbb{R}$ such that*

$$\|F_n\|_{L^p(\mathbb{R}; X)} \rightarrow \|f\|_{L^p(\mathbb{T}; X)}$$

and, for each j ,

$$\left\| D_{I_{j,n}}^{\mathbb{R}} F_n \right\|_{L^p(\mathbb{R}; X)} \rightarrow \left\| D_{I_j}^{\mathbb{T}} f \right\|_{L^p(\mathbb{T}; X)}.$$

Consequently the same convergence holds for the finite $\ell^q(L^p)$ -norms of the tuples of interval projections.

Proof. This is the usual proof of de Leeuw's compactification and restriction theorems for L^p -multipliers, applied to the finite tuple of symbols $(\mathbf{1}_{J_j})_{j=1}^N$ or $(\mathbf{1}_{I_j})_{j=1}^N$. We recall the points relevant to the mixed norm.

For compactification, sample the real-line multiplier at the lattice $\varepsilon\mathbb{Z}$. The preimages

$$J_{j,\varepsilon} := \{k \in \mathbb{Z} : \varepsilon k \in J_j\}$$

are intervals in $\mathbb{Z}$, and they are disjoint when the J_j 's are disjoint. The standard periodization argument, with the usual normalization of dilates, gives

$$\|f_\varepsilon\|_{L^p(\mathbb{T}; X)} \rightarrow \|f\|_{L^p(\mathbb{R}; X)}$$

and, for each fixed j ,

$$\left\| D_{J_{j,\varepsilon}}^{\mathbb{T}} f_\varepsilon \right\|_{L^p(\mathbb{T}; X)} \rightarrow \left\| D_{J_j}^{\mathbb{R}} f \right\|_{L^p(\mathbb{R}; X)}.$$

The endpoint-avoidance assumption on $\widehat{f}$ removes the only discontinuity issue; a standard approximation removes it for general test functions.

For restriction, choose real intervals $\tilde{I}_j \subset \mathbb{R}$ with $\tilde{I}_j \cap \mathbb{Z} = I_j$, for instance by putting endpoints halfway between adjacent integers. The classical restriction proof localizes a rescaled copy of the

trigonometric polynomial on $\mathbb{R}$ so that its Fourier mass concentrates near the integer frequencies. Applying this to each $\tilde{I}_j$ yields the displayed componentwise convergence.

These are exactly de Leeuw's arguments [2]; see also the operator-valued presentation in [3, Chapter 5]. Since the family has only $N < \infty$ components, componentwise convergence of the L^p -norms implies convergence of the finite ℓ^q -norm, including $q = \infty$. $\square$

Proof of the theorem. We first prove the lower estimate. Suppose X has lower $\ell^q(L^p)$ -decompositions on $\mathbb{T}$ with constant L . Let $\mathcal{J} = \{J_1, \dots, J_N\}$ be a finite disjoint interval family in $\mathbb{R}$, and let $f \in \mathcal{S}(\mathbb{R}; X)$ have compact Fourier support. By a harmless approximation we may assume that $\hat{f}$ does not charge the endpoints of the J_j 's. Apply Lemma 1(1). For every n ,

$$\left\| A_{\mathcal{J}_n}^{\mathbb{T}} f_n \right\|_{\ell^q(L^p(\mathbb{T}; X))} \leq L \|f_n\|_{L^p(\mathbb{T}; X)}.$$

Passing to the limit gives

$$\left\| A_{\mathcal{J}}^{\mathbb{R}} f \right\|_{\ell^q(L^p(\mathbb{R}; X))} \leq L \|f\|_{L^p(\mathbb{R}; X)}.$$

Density and monotone convergence over finite subfamilies give the real-line lower decomposition property.

Conversely, suppose the lower estimate holds on $\mathbb{R}$ with constant L . Let $\mathcal{I} = \{I_1, \dots, I_N\}$ be a finite disjoint interval family in $\mathbb{Z}$, and let $f \in \mathcal{P}(\mathbb{T}; X)$. Lemma 1(2) gives real-line approximants F_n and interval families $\mathcal{I}_n$. The real-line estimate gives

$$\left\| A_{\mathcal{I}_n}^{\mathbb{R}} F_n \right\|_{\ell^q(L^p(\mathbb{R}; X))} \leq L \|F_n\|_{L^p(\mathbb{R}; X)}.$$

Passing to the limit yields the corresponding estimate on $\mathbb{T}$. Thus the best lower constants on $\mathbb{T}$ and $\mathbb{R}$ are equal.

For upper decompositions, first take a finite interval family containing the Fourier support of the test function; equivalently, take the finite part of an interval partition on which the test function has nonzero projections. The same approximation lemma preserves this support condition: the preimage intervals contain the Fourier support of the approximating trigonometric polynomial, and in the restriction direction the thickened real intervals contain the Fourier support of the real-line approximants. The proof is then identical, with the inequality reversed. If the upper estimate holds on $\mathbb{T}$, then for the approximants in Lemma 1(1),

$$\|f_n\|_{L^p(\mathbb{T}; X)} \leq U \left\| A_{\mathcal{J}_n}^{\mathbb{T}} f_n \right\|_{\ell^q(L^p(\mathbb{T}; X))},$$

and the limit gives the real-line upper estimate. If the upper estimate holds on $\mathbb{R}$, Lemma 1(2) transfers it back to $\mathbb{T}$. The best upper constants are therefore equal as well. $\square$

Verification Notes

No numerical verification is involved. The proof reduces the open problem to the standard de Leeuw compactification/restriction limit and checks that the finite mixed norm used in Deng–Lorist–Veraar's decomposition definition is preserved under that limit. The argument is insensitive to the precise choice of open, closed, or half-open interval endpoints, because the estimates are first proved for test functions whose Fourier supports avoid endpoints and then extended by approximation.

Novelty and Literature Check

The run indexes were searched for arXiv:2311.01436, “Strongly Kreiss”, “UMD”, “Fourier decomposition”, “decomposition properties on $\mathbb{R}$ ”, and related torus/real-line transference phrases. No existing packet or attempt addresses this problem.

External web searches for exact phrases around the paper title, Problem 5.2, “ $\ell^q(L^p)$ -decompositions”, and torus/real-line decomposition transference found the source paper and adjacent multiplier literature, but no later paper explicitly resolving this open problem. Novelty confidence is moderate: the proof is a direct application of standard de Leeuw machinery once the mixed-norm finite-family formulation is isolated.

Limitations

The theorem is stated in the nontrivial L^p range $1 < p < \infty$, which is the range used in the source paper’s UMD/Fourier-multiplier theory. Endpoint conventions for $p = 1$ or $p = \infty$ are not addressed here. The packet does not solve the scalar endpoint Problem 5.3 or the sharp extrapolation Problem 5.4.

Human Review Recommendation

Reviewers should check Lemma 1: in particular, that the standard de Leeuw compactification and restriction proofs may be applied componentwise to finite interval families and that the convergence of component L^p -norms is enough to pass to the finite $\ell^q(L^p)$ -mixed norm without a cardinality-dependent loss.

References

- [1] C. Deng, E. Lorist, and M. C. Veraar, *Strongly Kreiss Bounded Operators in UMD Banach Spaces*, arXiv:2311.01436.
- [2] K. de Leeuw, *On L_p multipliers*, Ann. of Math. (2) **81** (1965), 364–379.
- [3] T. Hytönen, J. van Neerven, M. Veraar, and L. Weis, *Analysis in Banach Spaces. Volume I: Martingales and Littlewood–Paley Theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete 63, Springer, 2016.